

The background of the slide features a blue geometric pattern consisting of overlapping triangles and lines, creating a modern, abstract design. The pattern is more prominent at the top and bottom edges of the slide.

CSA0708

COMPUTER NETWORKS

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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB = 150,000,000 bytes**
- Bandwidth = **50 Mbps = 50,000,000 bits/s**

- Segment size = **1500 bytes**
- Data Link Layer overhead = **40 bytes/frame**

1. Number of segments

$$\frac{150,000,000}{1500} = 100,000 \text{ segments}$$

2. Number of frames

Each segment is encapsulated into one frame.

100,000 frames

3. Total Data Link Layer overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

4. Transmission time

Total data transmitted:

$$150 \text{ MB} + 4 \text{ MB} = 154 \text{ MB}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

$$\frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds}$$

Final Answer

- Segments = 100,000
- Frames = 100,000
- Data Link Layer overhead = 4 MB
- Transmission time = 24.64 seconds

Explanation:

The **Transport Layer** divides the file into segments, assigns sequence numbers, and ensures reliable delivery using acknowledgements and retransmissions. The **Data Link Layer** converts segments into frames, adds headers and trailers, detects errors using CRC/FCS, and provides reliable communication between directly connected devices.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Data per frame = 1024 bytes

1. Number of transmission windows

$$\frac{48}{6} = 8 \text{ windows}$$

2. Total retransmissions

One frame is lost in each window.

$$8 \times 1 = 8 \text{ retransmissions}$$

Final Answer

- Transmission windows = 8
- Retransmissions = 8 frames

Explanation:

Flow control regulates the rate of data transmission between the sender and receiver. It prevents buffer overflow, reduces congestion, and improves communication efficiency by ensuring the sender transmits data only as fast as the receiver can process it.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = **12,000 bytes**
- MTU = **1500 bytes**
- IP header per fragment = **20 bytes**

1. Number of fragments

Maximum data per fragment:

$$1500 - 20 = 1480 \text{ bytes}$$

$$\frac{12,000}{1480} \approx 8.1$$

So, **9 fragments** are required.

2. Total header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Final Answer

- Number of fragments = 9
- Total IP header overhead = 180 bytes

Explanation:

The **Network Layer** fragments large IP packets to fit the MTU of the network. Each fragment carries its own IP header with identification, fragment offset, and more-fragments information. At the destination, these fragments are reassembled to reconstruct the original packet, ensuring successful delivery.

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$$\frac{48}{6} = 8 \text{ windows}$$

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One frame is lost in each window.

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Final Answer

- Transmission windows = 8
- Retransmissions = 8 frames

Explanation:

Flow control regulates the data transmission rate between the sender and receiver. It prevents buffer overflow, reduces congestion, and improves communication efficiency by ensuring the sender transmits data only at a rate the receiver can handle.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video file size = 600 MB
- Checkpoint after every 60 MB
- Failure occurs after 390 MB

1. Number of checkpoints before failure

$$\frac{390}{60} = 6 \text{ checkpoints}$$

(Checkpoints are at 60, 120, 180, 240, 300, and 360 MB.)

2. Data to be retransmitted

The last checkpoint was at 360 MB.

$$390 - 360 = 30 \text{ MB}$$

Final Answer

- Checkpoints created before failure = 6
- Data to be retransmitted = 30 MB

Explanation:

The Session Layer uses synchronization checkpoints to save transmission progress. If a failure occurs, data transfer resumes from the last checkpoint instead of starting over, reducing retransmission time and improving communication reliability.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5% of compressed size

1. Compressed file size

$$900 - (40\% \times 900) = 900 - 360 = 540 \text{ MB}$$

2. Encrypted file size

$$540 + (5\% \times 540) = 540 + 27 = 567 \text{ MB}$$

3. Total percentage reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Final Answer

- Compressed file size = 540 MB
- Encrypted file size = 567 MB
- Total percentage reduction = 37%

Explanation:

The Presentation Layer improves transmission efficiency by compressing data, reducing the amount of data sent. It also enhances security by encrypting the compressed data, protecting it from unauthorized access during transmission.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Data size = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

1. Total number of requests

$$\frac{3000}{3} = 1000 \text{ requests}$$

2. Transmission time

Convert data to bits:

$$3 \text{ GB} = 3000 \text{ MB} = 24,000 \text{ Mb}$$

$$\text{Transmission Time} = \frac{24,000}{120} = 200 \text{ seconds}$$

Final Answer

- Total requests = 1000
- Transmission time = 200 seconds

Explanation:

The Application Layer provides services such as file transfer (FTP), manages user requests, ensures reliable data exchange, and allows users to access network resources efficiently.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application Layer delay = 4 ms
- Transport Layer delay = 3 ms
- Network Layer delay = 5 ms
- Data Link Layer delay = 2 ms
- Physical Layer delay = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

1. Total End-to-End Delay

$$4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$$

Final Answer

- Total end-to-end delay = 45 ms

Explanation:

Each OSI layer performs a specific function:

- Application Layer: Provides network services to applications.
- Transport Layer: Ensures reliable data delivery.
- Network Layer: Handles routing and packet forwarding.

- Data Link Layer: Provides framing and error detection.
- Physical Layer: Transmits data as signals over the medium.

Together, these layers ensure efficient and reliable communication across the network.

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Protocol overhead per frame = 80 bytes

1. Number of frames

$$\frac{80,000,000}{1600} = 50,000 \text{ frames}$$

2. Total protocol overhead

$$50,000 \times 80 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

3. Transmission efficiency

Total data transmitted:

$$80 + 4 = 84 \text{ MB}$$

$$\text{Efficiency} = \frac{80}{84} \times 100 = 95.24\%$$

Final Answer

- Number of frames = 50,000
- Total protocol overhead = 4 MB
- Transmission efficiency = 95.24%

Explanation:

Protocol overhead consists of headers and trailers added by different OSI layers. While it enables functions such as addressing, error detection, and reliable communication, excessive overhead reduces the effective data transmission efficiency and consumes additional bandwidth.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Medical record size = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link Layer overhead = 60 bytes/frame

1. Compressed file size

$$240 - (30\% \times 240) = 240 - 72 = 168 \text{ MB}$$

2. Number of segments

Convert compressed size to bytes:

$$168 \text{ MB} = 168,000,000 \text{ bytes}$$

$$\frac{168,000,000}{1400} = 120,000 \text{ segments}$$

3. Total protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

4. Final data transmitted

$$168 + 7.2 = 175.2 \text{ MB}$$

Final Answer

- Compressed file size = 168 MB
- Number of segments = 120,000
- Total protocol overhead = 7.2 MB
- Final data transmitted = 175.2 MB

Explanation:

The Presentation Layer compresses and encrypts data for efficiency and security. The Transport Layer divides the data into segments for reliable delivery. The Data Link Layer adds headers and trailers for framing and error detection, while the Physical Layer transmits the final data over the network, ensuring secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand